

GDPS Metro Global - GM Workshop Scripts and Procedures



GDPS Metro script statements for MGM



- Unless otherwise stated, all 2-site GDPS Metro script statements can also be used in a GDPS MGM configuration except:
 - Use DASD 'RECOVER SIMPLEX' with caution (only if really required...)
- Only valid in an MGM IR configuration:
 - **DASD 'INCRESYN SETGM'**
 - Set IR on the current PPRC primary disk
 - Only allowed when the GDPS Metro DASD mirroring status is green OK
 - Briefly sets mirroring status to NOK and performs a suspend/resume of the PPRC pairs
 - Use Monitor2 or DASD START SECONDARY to reenable HyperSwap
 - Wait at least 15 minutes after completion and verify GM consistency group formation
 - **DASD 'INCRESYN RESETGM'**
 - Clear IR from the current PPRC primary disk
 - Only allowed when the GDPS Metro mirroring status is green OK
 - Briefly sets mirroring status to NOK and performs a suspend/resume of the PPRC pairs
 - Use Monitor2 or DASD START SECONDARY to reenable HyperSwap

GDPS Metro script statements for MGM



- Use case for manually setting incremental resync:
 - There are only limited use cases when you will need to use the INCRESYN commands:
 - When adding devices to the CONFIG that have already been made duplex outside of GDPS
 - Removing devices from the configuration so clearing the status
- HyperSwap and IR
 - When running in a cascaded mode (RegA.RS1 to RegA.RS2 to RegB.RS1) GDPS needs to use the MGM IR bitmap on the RS1 disk to allow for MOVEGM processing in the event of a loss of the RegA.RS2 disk.
 - This is accomplished by START SECONDARY processing or through use of the INCRESYN SETGM command
 - HyperSwap to RS2, and GDPS will use the Multi-Target IR bitmap as the RS2 disk is now primary for Metro and GM
 - Following a START SECONDARY (which cleans up the old MGM IR settings) and subsequent HyperSwap back to RS1 and START SECONDARY to return to the original cascaded mode, MGM IR will be set on by the START SECONDARY script command.
 - You still have to wait for the 15 mins to ensure MGM IR is fully initialized.

MGM processing for GDASD script statements



- Existing 2-site GDPS GM GDASD statements work mostly the same – based on the GM Site indicator
 - Using ALL for session name parameter of GDASD statements is not allowed
 - Only a single session is allowed in MGM
 - Some additional processing relating to IR
- **GDASD 'sessname CHGGMSITE'**
 - Will flip the GM Site indicator between RegA.RS2 and RegA.RS1 if needed – this defines which RegA disk the GM Site indicator is set to and hence the intended orientation of the GM session.
- **GDASD 'sessname INITIALIZE'**
 - Standard 2-site processing: Initialize the RegA.RS2 to RegB.RS1 GM mirror and additionally:
 - Establish RegA.RS1 to RegB.RS1 paths
 - Prepare, in the RegA.RS1 disk, GM “topology” in preparation for an IR operation:
 - Define the RegA.RS1 disk LSSs to the GM session in the RegA.RS1 disk
 - Join the RegA.RS1 disk devices to the GM session in the RegA.RS1 disk
- **GDASD 'sessname CLEANUP'**
 - Clean up the RegA.RS2 to RegB.RS1 GM mirror and additionally:
 - Clean up (undo) the IR related processing performed by INITIALIZE
 - Does not remove the MGM IR bitmaps from the RegA.RS1 disk (see INCRESYN command)

Note: although the above talk about the RegA.RS2 to RegB.RS1 *GM mirror*, the statements will also work for INITIALIZE or CLEANUP of RegA.RS1 to RegB.RS1 mirror if running GM from RegA.RS1 to RegB.RS1

Other script statements for use in MGM IR



- **REMOTE** '{*alias* | *aliasgroup*} **SCRIPT**=*scriptname* [**WTIME**=*hh:mm:ss*]'
 - Execute a script on a remote target system as identified by alias/aliasgroup
 - *aliasgroup* is resolved to a single target at execution time
 - Scriptname must exist in the remote target(s)
 - Script to be executed remotely must not include a REMOTE or PROCEDURE statement
 - Avoid deadlocks
 - WTIME (optional) timeout
 - Source system waits this long for target system to signal completion
 - Default 30 mins if not coded
 - REMOTE statement fails with GEO117/118 on source system if not completed
 - Script continues execution on target even if REMOTE statement is failed
 - Pre execution validation – cannot initiate script with a REMOTE statement if target system is executing a script

Other script statements for use in MGM IR



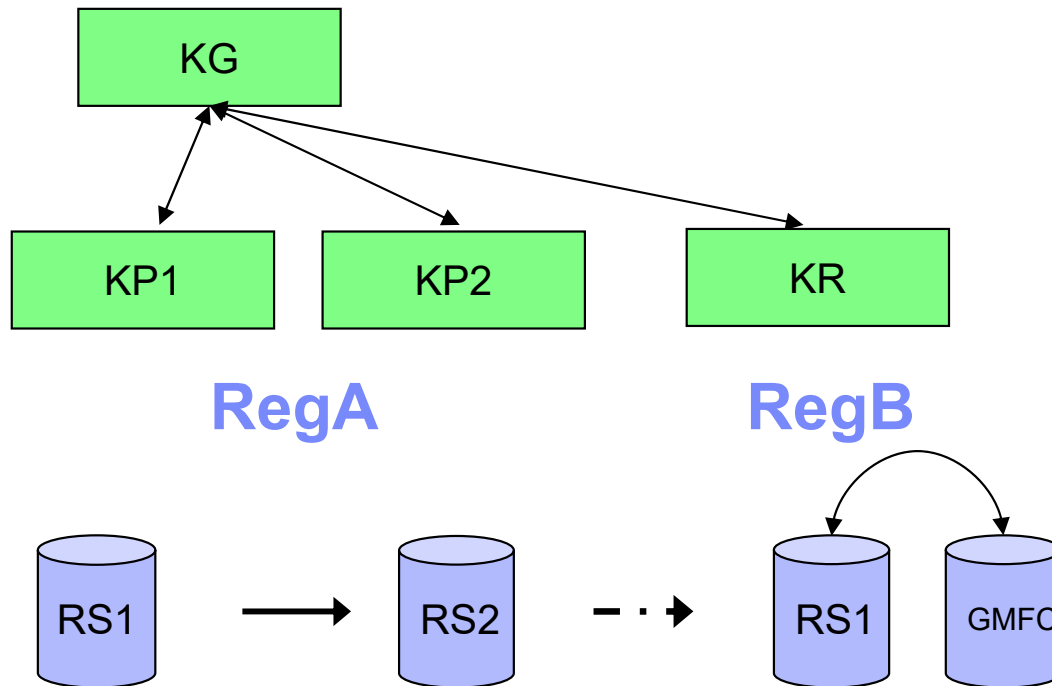
- **PROCEDURE='local_sessname/remote_sessname proc_name [PROMPT|NOPROMPT]'**
 - Kg-sys only
 - *Local_sessname is mandatory and specifies the GM session name in the local Kg system executing the procedure statement*
 - *Remote_sessname is required in MGM 4-Site and specifies the GM session in the opposing region Kg-system*
 - PROMPT issues prompts to the user during specific execution steps
 - Recommendation: Always use the PROMPT option
 - PROCEDURE statement can not be included in a remotely executed script

GDPS PROCEDURES - Overview



- GDPS PROCEDURE script statement
 - Executes GDPS-supplied “Procedures”
 - Facilitates most MGM 3-site (and 4-site) processing scenarios
 - GDPS ships sample GEOGSCR4 in the SGDPSAMP library
- GDPS/MGM 4-site IR configuration PROCEDURES are supplied for complex scenarios such as:
 - Planned HyperSwap (RS1 to RS2 or RS2 to RS1 in the Metro region)
 - Reintroduction of intermediate disk (eg: following RS2 failure in the Metro region)
 - Planned region switch
 - Return home after planned region switch
 - Return home after unplanned recovery in Region B
- PROCEDURES coordinate execution across multiple K-systems
 - Using GEOGROUP information
- PROCEDURES provides restart capability after a failed execution
 - Fix the problem before re-execution of the PROCEDURE
 - Re-execute the same procedure
 - Steps previously executed successfully are skipped
 - Execution restarts at an appropriate point
 - Confirmation prompt is issued for restarted procedure with PROMPT option
- PROCEDURES are aware of the current orientation of the replication

Procedure execution – remote function requests



- Procedure is always initiated on the Kg NetView
 - A Procedure can perform *remote function calls* to other k-sys using GEOGROUP info for communication .
 - For example:
 - HyperSwap can only be performed on the Master Kp
 - RegB disk actions must be carried out by Kr
 - Success/failure of remote function is reported back to Kg
 - All Steps logged in SDF & Netlog

- A hypothetical Procedure execution
 - Kg initiates execution
 - Kp: check Metro status
 - Kr: check GM Secondary status
 - Kg: check GM relationship
 - Kr: change RegB.RS1 to RegA.RS1 relationship
 - Kp: change RegA.RS1 to RegA.RS2 relationship
 - Kp: check status
 - etc



SDF trace entries for a sample PROCEDURE execution



```
SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:41:03

----- Trace Entries ----- 1/16(57)

Time      Text
17:40:04  MGM4_SWAP PLANNED/STANDARD ACTION STARTED FROM STEP 1
17:40:04  PROCEDURE=ZPROD01/ZPROD01 MGM4SITE_SWAP STARTED
17:40:04  ESTABLISHING CONTACT WITH TARGET SYSTEMS REQUIRED BY MGM4SITE_SWAP
17:40:05  PROCEDURE MGM4SITE_SWAPMTFO_CAS_RS2 WILL BE SCHEDULED
17:40:05  SUCCESSFULLY CONTACTED ALL TARGET SYSTEMS REQUIRED BY MGM4SITE_SWAPMT
17:40:05  GEOGROUP VALIDATION SUCCESSFUL IN ALL TARGET SYSTEMS REQUIRED BY MGM4
17:40:06  CONFIGURATIONS SUCCESSFULLY CREATED IN ALL TARGET SYSTEMS REQUIRED BY
17:40:06  PROCEDURE VERSION IS PH39005 BUILT 22161 AND CONSISTS OF 57 STEPS
17:40:06  MGM4SITE_SWAPMTFO_CAS_RS2 PROCEDURE EXECUTION STARTED
17:40:06  STEP(001) MGM4SITE PLANNED HYPERSWAP TO MOP.ANTIGONE VOLUMES
17:40:06  STEP(001) VERIFYING VOLUMES IN REGIONA
17:40:06  STEP(002) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(CHECK) STARTED
17:40:07  STEP(002) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(CHECK) ENDED RC=00
17:40:07  STEP(003) MOP.KP.A6PS1 FUNCTION(PQUERY) STARTED
17:40:08  STEP(003) MOP.KP.A6PS1 FUNCTION(PQUERY) ENDED RC=00
17:40:08  STEP(004) MOP.KP.A6PS1 FUNCTION(PQUERY) STARTED

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
```

SDF trace entries for a sample PROCEDURE execution



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SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:42:36

----- Trace Entries ----- 17/32(79)
Time      Text
17:40:09  STEP(004) MOP.KP.A6PS1 FUNCTION(PQUERY) ENDED RC=00
17:40:09  STEP(005) MOP.KG.A6QS1 FUNCTION(PQUERY) STARTED
17:40:10  STEP(005) MOP.KG.A6QS1 FUNCTION(PQUERY) ENDED RC=00
17:40:10  STEP(006) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
17:40:10  STEP(006) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED RC=00
17:40:10  STEP(009) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
17:40:12  STEP(009) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
17:40:12  STEP(010) REGIONA VOLUME VERIFICATION COMPLETE
17:40:12  STEP(010) PROCEEDING WITH PLANNED HYPERSWAP IN REGIONA
17:40:12  STEP(011) MOP.KP.A6PS1 FUNCTION(SITE) ACTION(CHANG) SITE(SITE2) STAR
17:40:13  STEP(011) MOP.KP.A6PS1 FUNCTION(SITE) ACTION(CHANG) SITE(SITE2) ENDE
17:40:13  STEP(012) MOP.KP.A6PS1 FUNCTION(GDPS) COMMAND(SWAPMIRROR) STARTED
17:40:17  STEP(012) MOP.KP.A6PS1 FUNCTION(GDPS) COMMAND(SWAPMIRROR) ENDED RC=00
17:40:17  STEP(013) REGIONA HA PROTECTION DISABLED
17:40:17  STEP(014) MOP.KP.A6PS1 FUNCTION(SOFTFENCE) STARTED
17:40:19  STEP(014) MOP.KP.A6PS1 FUNCTION(SOFTFENCE) ENDED RC=00

==>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
  
```

SDF trace entries for a sample PROCEDURE execution



```
SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:43:43

----- Trace Entries ----- 33/48(114)

Time      Text
17:40:19  STEP(015) MOP.KP.A6PS1 FUNCTION(THAW) STARTED
17:40:20  STEP(015) MOP.KP.A6PS1 FUNCTION(THAW) ENDED RC=00
17:40:20  STEP(016) PLANNED HYPERSWAP TO MOP.ANTIGONE VOLUMES COMPLETE
17:40:20  STEP(016) RESTORING MOP.ANTIGONE --> MOP.ODYSSEUM MM VOLUMES
17:40:20  STEP(019) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
17:40:20  STEP(019) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED RC=00
17:40:20  STEP(020) MOP.KP.A6PS1 FUNCTION(PESTPAIR) STARTED
17:40:22  STEP(020) MOP.KP.A6PS1 FUNCTION(PESTPAIR) ENDED RC=00
17:40:22  STEP(021) MOP.KG.A6QS1 FUNCTION(PESTPAIR) STARTED
17:40:25  STEP(021) MOP.KG.A6QS1 FUNCTION(PESTPAIR) ENDED RC=00
17:40:25  STEP(022) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(DUPLEX) STARTED
17:40:26      (022) ALL MONITORED VOLUMES ARE NOW DUPLEX
17:40:26  STEP(022) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(DUPLEX) ENDED RC=00
17:40:26  STEP(023) MOP.KP.A6PS1 FUNCTION(PESTPAIR) STARTED
17:40:28  STEP(023) MOP.KP.A6PS1 FUNCTION(PESTPAIR) ENDED RC=00
17:40:28  STEP(024) MOP.KP.A6PS1 FUNCTION(MONITOR) CRITERIA(DUPLEX) STARTED

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
21=DTRACE 22=DDASD 23=DAUT
```

SDF trace entries for a sample PROCEDURE execution



```

SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:44:45

----- Trace Entries ----- 49/64(114)

Time      Text
17:40:29  (024) ALL MONITORED VOLUMES ARE NOW DUPLEX
17:40:29  STEP(024) MOP.KP.A6PS1 FUNCTION(MONITOR) CRITERIA(DUPLEX) ENDED RC=00
17:40:29  STEP(025) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(YES) STARTED
17:40:29  STEP(025) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(YES) ENDED RC=00
17:40:29  STEP(026) MOP.KP.A6PS1 FUNCTION(MONITOR2) STARTED
17:40:32  STEP(026) MOP.KP.A6PS1 FUNCTION(MONITOR2) ENDED RC=00
17:40:32  STEP(027) REGIONA HA PROTECTION ENABLED
17:40:32  STEP(028) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(OOS) STARTED
17:42:04  (028) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET 99.500
17:42:04  STEP(028) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(OOS) ENDED RC=00
17:42:04  STEP(029) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(OOS) STARTED
17:42:23  (029) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET 99.500
17:42:23  STEP(029) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(OOS) ENDED RC=00
17:42:23  STEP(030) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) STARTED
17:42:24  STEP(030) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) ENDED RC=00
17:42:24  STEP(031) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
  
```

SDF trace entries for a sample PROCEDURE execution



```
SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:45:29
----- Trace Entries ----- 65/80(114)
Time      Text
17:42:27  STEP(031) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) ENDED RC=00
17:42:27  STEP(032) REGIONB DR COPY AGING
17:42:27  STEP(033) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
17:42:29  STEP(033) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
17:42:29  STEP(034) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANG) SITE(SITE2) STAR
17:42:30  STEP(034) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANG) SITE(SITE2) ENDE
17:42:30  STEP(036) USA.KR.A6RS4 FUNCTION(PESTPAIR) STARTED
17:42:32  STEP(036) USA.KR.A6RS4 FUNCTION(PESTPAIR) ENDED RC=00
17:42:32  STEP(037) MOP.KG.A6QS1 FUNCTION(PESTPAIR) STARTED
17:42:34  STEP(037) MOP.KG.A6QS1 FUNCTION(PESTPAIR) ENDED RC=00
17:42:34  STEP(038) INCREMENTAL COPY MOP.ODYSSEUM --> USA.MANYCOVE STARTED
17:42:34  STEP(039) CONDITIONALLY SUSPEND/RESYNC MOP.ODYSSEUM --> USA.MANYCOVE
17:42:34  STEP(040) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
17:42:36  STEP(040) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
17:42:36  STEP(041) MOP.KG.A6QS1 FUNCTION(PESTPAIR) STARTED
17:42:38  STEP(041) MOP.KG.A6QS1 FUNCTION(PESTPAIR) ENDED RC=00

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
```

SDF trace entries for a sample PROCEDURE execution



```

SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:46:10

----- Trace Entries ----- 81/96(114)

Time      Text
17:42:38  STEP(042) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
17:42:38  STEP(042) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED RC=00
17:42:38  STEP(043) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) STARTED
17:42:38  STEP(043) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) ENDED RC=00
17:42:38  STEP(044) MOP.KG.A6QS1 FUNCTION(PDELPAIR) STARTED
17:42:40  STEP(044) MOP.KG.A6QS1 FUNCTION(PDELPAIR) ENDED RC=00
17:42:40  STEP(045) USA.KR.A6RS4 FUNCTION(PDELPAIR) STARTED
17:42:43  STEP(045) USA.KR.A6RS4 FUNCTION(PDELPAIR) ENDED RC=00
17:42:43  STEP(046) USA.KG.A6QS4 FUNCTION(PDELPAIR) STARTED
17:42:45  STEP(046) USA.KG.A6QS4 FUNCTION(PDELPAIR) ENDED RC=00
17:42:45  STEP(047) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(FIRSTPASS) STARTED
17:42:46      (047) ALL MONITORED VOLUMES HAVE NOW COMPLETED FIRSTPASS
17:42:46  STEP(047) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(FIRSTPASS) ENDED RC
17:42:46  STEP(049) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANGE) SITE(SITE2) STAR
17:42:47  STEP(049) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANGE) SITE(SITE2) ENDE
17:42:47  STEP(050) MOP.KG.A6QS1 FUNCTION(GMSTART) SITE(SITE2) STARTED

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
  
```

SDF trace entries for a sample PROCEDURE execution



```
SDF - G0C1GDPS      GDPS Disaster/Recovery Readiness      09/12/23 17:46:50

----- Trace Entries ----- 97/112(114)

Time      Text
17:42:50  STEP(050) MOP.KG.A6QS1 FUNCTION(GMSTART) SITE(SITE2) ENDED RC=00
17:42:50  STEP(051) REGIONB DR COPY RESTARTED
17:42:50  STEP(052) USA.KG.A6QS4 FUNCTION(HMTCTRL) STARTED
17:42:51  STEP(052) USA.KG.A6QS4 FUNCTION(HMTCTRL) ENDED RC=00
17:42:51  STEP(053) MOP.KP.A6PS1 FUNCTION(HMTCTRL) STARTED
17:42:52  STEP(053) MOP.KP.A6PS1 FUNCTION(HMTCTRL) ENDED RC=00
17:42:52  STEP(054) MOP.KP.A6PS1 FUNCTION(HMTCTRL) STARTED
17:42:53  STEP(054) MOP.KP.A6PS1 FUNCTION(HMTCTRL) ENDED RC=00
17:42:53  STEP(055) MOP.KG.A6QS1 FUNCTION(HMTCTRL) STARTED
17:42:54  STEP(055) MOP.KG.A6QS1 FUNCTION(HMTCTRL) ENDED RC=00
17:42:54  STEP(056) MOP.KG.A6QS1 FUNCTION(HMTCTRL) STARTED
17:42:55  STEP(056) MOP.KG.A6QS1 FUNCTION(HMTCTRL) ENDED RC=00
17:42:55  STEP(057) MGM4SITE PLANNED HYPERSWAP TO MOP.ANTIGONE VOLUMES COMPLETE
17:42:55  PROCEDURE EXECUTION SUCCESSFUL RC=00
17:42:55  PROCEDURE=ZPROD01/ZPROD01 MGM4SITE_SWAP ENDED RC=0
17:42:55  MGM4_SWAP PLANNED/STANDARD ACTION ENDED

===>
1=Help 2=Detail 3=Return      6=Roll      9=Bottom 10=Previous 11=Next 12=Top
                                21=DTRACE 22=DDASD 23=DAUT
```


PROCEDURE execution Summary Report



```

*****
GEORPT * - START OF PROCEDURE EXECUTION LOG
GEORPT
*****
GEORPT * 16:34:13 ESTABLISHING CONTACT WITH TARGET SYSTEMS REQUIRED BY MGM4SITE_RESTORE
GEORPT * 16:34:14 QUERYING STATE OF MM PRIMARY VOLUMES IN REGIONA REGION
GEORPT * 16:34:15 PROCEDURE MGM4SITE_RESTORE_MT_RS2 WILL BE SCHEDULED
GEORPT * 16:34:15 SUCCESSFULLY CONTACTED ALL TARGET SYSTEMS REQUIRED BY
MGM4SITE_RESTORE_MT_RS2
GEORPT * 16:34:15 GEOGROUP VALIDATION SUCCESSFUL IN ALL TARGET SYSTEMS REQUIRED
BY MGM4SITE_RESTORE_MT_RS2
GEORPT * 16:34:16 CONFIGURATIONS SUCCESSFULLY CREATED IN ALL TARGET SYSTEMS
REQUIRED BY MGM4SITE_RESTORE_MT_RS2
GEORPT * 16:34:16 PROCEDURE VERSION IS PH44166 BUILT 22231 AND CONSISTS OF 50
STEPS
GEORPT * 16:34:16 MGM4SITE_RESTORE_MT_RS2 PROCEDURE EXECUTION STARTED
GEORPT * 16:34:16 STEP(001) MOP.KG.A6QS1 FUNCTION(PQUERY) STARTED
GEORPT * 16:34:17 STEP(001) MOP.KG.A6QS1 FUNCTION(PQUERY) ENDED RC=00
PRINT LOG/TRACE UTILITY 09/12/23 56
GEORPT * 16:34:17 STEP(002) MOP.KP.A6PS1 FUNCTION(PQUERY) STARTED
GEORPT * 16:34:18 STEP(002) MOP.KP.A6PS1 FUNCTION(PQUERY) ENDED RC=00
GEORPT * 16:34:18 STEP(003) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
GEORPT * 16:34:20 STEP(003) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
GEORPT * 16:35:20 STEP(004) MOP.KG.A6QS1 FUNCTION(PQUERY) STARTED
GEORPT * 16:35:21 STEP(004) MOP.KG.A6QS1 FUNCTION(PQUERY) ENDED RC=00
GEORPT * 16:35:21 STEP(005) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
GEORPT * 16:35:21 STEP(005) MOP.KG.A6QS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED
RC=00
GEORPT * 16:35:21 STEP(006) MOP.KG.A6QS1 FUNCTION(PQUERY) BYPASSED - NO
COPYONCE VOLUMES
GEORPT * 16:35:21 STEP(007) MOP.KG.A6QS1 FUNCTION(PESTPAIR) BYPASSED - NO
COPYONCE VOLUMES
GEORPT * 16:35:21 STEP(008) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(FIRSTPASS)
BYPASSED - NO COPYONCE VOLUMES
GEORPT * 16:35:21 STEP(009) MOP.KG.A6QS1 FUNCTION(PQUERY) STARTED
GEORPT * 16:35:22 STEP(009) MOP.KG.A6QS1 FUNCTION(PQUERY) ENDED RC=00
GEORPT * 16:35:22 STEP(010) USA.KR.A6RS4 FUNCTION(PQUERY) STARTED
GEORPT * 16:35:23 STEP(010) USA.KR.A6RS4 FUNCTION(PQUERY) ENDED RC=00

```

PROCEDURE execution Summary Report



```

GEORPT * 16:35:23  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24      (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:24  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:24  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:25      (012) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:25  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:25  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED
GEORPT * 16:35:26  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) ENDED
RC=00
GEORPT * 16:35:26  STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) STARTED
GEORPT * 16:35:29  STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) ENDED
RC=00
GEORPT * 16:35:29  STEP(015) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:35:29      REGIONB DR COPY AGING
GEORPT * 16:35:29  STEP(015) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:35:29  STEP(016) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
GEORPT * 16:35:31  STEP(016) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
GEORPT * 16:35:31  STEP(017) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) STARTED
GEORPT * 16:35:31  STEP(017) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) ENDED
RC=00
GEORPT * 16:35:31  STEP(018) USA.KR.A6RS4 FUNCTION(PESTPAIR) STARTED
GEORPT * 16:35:33  STEP(018) USA.KR.A6RS4 FUNCTION(PESTPAIR) ENDED RC=00
GEORPT * 16:35:33  STEP(019) USA.KR.A6RS4 FUNCTION(PQUERY) BYPASSED -
PRIORFAILURE(YES)
GEORPT * 16:35:33  STEP(020) USA.KR.A6RS4 FUNCTION(GMRECOVER) STARTED
GEORPT * 16:35:44  STEP(020) USA.KR.A6RS4 FUNCTION(GMRECOVER) ENDED RC=00
GEORPT * 16:35:44  STEP(021) MOP.KP.A6PS1 FUNCTION(PQUERY) BYPASSED -
PRIORFAILURE(YES)
GEORPT * 16:35:44  STEP(022) MOP.KP.A6PS1 FUNCTION(PQUERY) BYPASSED -

```

PROCEDURE execution Summary Report



```

GEORPT * 16:35:23  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24      (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:24  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:24  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:25      (012) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:25  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:25  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED
GEORPT * 16:35:26  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) ENDED
RC=00
GEORPT * 16:35:26  STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) STARTED
GEORPT * 16:35:29  STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) ENDED
RC=00
GEORPT * 16:35:29  STEP(015) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:35:29      REGIONB DR COPY AGING
GEORPT * 16:35:29  STEP(015) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:35:29  STEP(016) MOP.KG.A6QS1 FUNCTION(PSUSPEND) STARTED
GEORPT * 16:35:31  STEP(016) MOP.KG.A6QS1 FUNCTION(PSUSPEND) ENDED RC=00
GEORPT * 16:35:31  STEP(017) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) STARTED
GEORPT * 16:35:31  STEP(017) USA.KR.A6RS4 FUNCTION(PESTPATH) CGROUP(NO) ENDED
RC=00
GEORPT * 16:35:31  STEP(018) USA.KR.A6RS4 FUNCTION(PESTPAIR) STARTED
GEORPT * 16:35:33  STEP(018) USA.KR.A6RS4 FUNCTION(PESTPAIR) ENDED RC=00
GEORPT * 16:35:33  STEP(019) USA.KR.A6RS4 FUNCTION(PQUERY) BYPASSED -
PRIORFAILURE(YES)
GEORPT * 16:35:33  STEP(020) USA.KR.A6RS4 FUNCTION(GMRECOVER) STARTED
GEORPT * 16:35:44  STEP(020) USA.KR.A6RS4 FUNCTION(GMRECOVER) ENDED RC=00
GEORPT * 16:35:44  STEP(021) MOP.KP.A6PS1 FUNCTION(PQUERY) BYPASSED -
PRIORFAILURE(YES)
GEORPT * 16:35:44  STEP(022) MOP.KP.A6PS1 FUNCTION(PQUERY) BYPASSED -
PRIORFAILURE(YES)

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PROCEDURE execution Summary Report



```
GEORPT * 16:35:23 STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24 (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:24 STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:24 STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:25 (012) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:25 STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:25 STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED
GEORPT * 16:35:26 STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) ENDED
RC=00
GEORPT * 16:35:26 STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) STARTED
GEORPT * 16:35:29 STEP(014) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE2) ENDED
RC=00
GEORPT * 16:35:29 ST GEORPT * 16:35:44 STEP(023) USA.KR.A6RS4 FUNCTION(PQUERY) STARTED
GEORPT * 16:35:46 STEP(023) USA.KR.A6RS4 FUNCTION(PQUERY) ENDED RC=00
GEORPT * 16:35:46 STEP(024) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(OFF) STARTED
GEORPT * 16:35:47 STEP(024) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(OFF) ENDED
RC=00
GEORPT * 16:35:47 STEP(025) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:35:47 REGIONA HA PROTECTION DISABLED
GEORPT * 16:35:47 STEP(025) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:35:47 STEP(026) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(DEREGISTER)
STARTED
GEORPT * 16:35:48 STEP(026) MOP.KP.A6PS1 FUNCTION(HYPERSW) ACTION(DEREGISTER)
ENDED RC=00
GEORPT * 16:35:48 STEP(027) MOP.KP.A6PS1 FUNCTION(GDPS) COMMAND(STOPMIRROR)
STARTED
```

PROCEDURE execution Summary Report



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GEORPT * 16:35:23  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24      (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:24  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:24  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:25      (012) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:25  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:25  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED
GEORPT * 16:35:2 GEORPT * 16:35:50  STEP(027) MOP.KP.A6PS1 FUNCTION(GDPS) COMMAND(STOPMIRROR)
ENDED RC=00
GEORPT * 16:35:50  STEP(028) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
GEORPT * 16:35:50  STEP(028) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED
RC=00
GEORPT * 16:35:50  STEP(029) MOP.KG.A6QS1 FUNCTION(PESTPAIR) STARTED
GEORPT * 16:35:52  STEP(029) MOP.KG.A6QS1 FUNCTION(PESTPAIR) ENDED RC=00
GEORPT * 16:35:52  STEP(030) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(DUPLEX)
STARTED
GEORPT * 16:35:53      (030) ALL MONITORED VOLUMES ARE NOW DUPLEX
GEORPT * 16:35:53  STEP(030) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(DUPLEX)
ENDED RC=00
GEORPT * 16:35:53  STEP(031) MOP.KP.A6PS1 FUNCTION(PESTPAIR) STARTED
GEORPT * 16:35:55  STEP(031) MOP.KP.A6PS1 FUNCTION(PESTPAIR) ENDED RC=00
GEORPT * 16:35:55  STEP(032) MOP.KP.A6PS1 FUNCTION(MONITOR) CRITERIA(DUPLEX)
STARTED
GEORPT * 16:35:56      (032) ALL MONITORED VOLUMES ARE NOW DUPLEX
GEORPT * 16:35:56  STEP(032) MOP.KP.A6PS1 FUNCTION(MONITOR) CRITERIA(DUPLEX)
ENDED RC=00
GEORPT * 16:35:56  STEP(033) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(YES) STARTED
GEORPT * 16:35:56  STEP(033) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(YES) ENDED
RC=00
GEORPT * 16:35:56  STEP(034) MOP.KP.A6PS1 FUNCTION(MONITOR2) STARTED
GEORPT * 16:35:59  STEP(034) MOP.KP.A6PS1 FUNCTION(MONITOR2) ENDED RC=00
GEORPT * 16:35:59  STEP(035) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:35:59      REGIONA HA PROTECTION ENABLED
GEORPT * 16:35:59  STEP(035) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:35:59  STEP(036) MOP.KG.A6QS1 FUNCTION(PESTPAIR) STARTED

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PROCEDURE execution Summary Report



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GEORPT * 16:35:23  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24      (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:24  STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:24  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:25      (012) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500
GEORPT * 16:35:25  STEP(012) USA.KR.A6RS4 FUNCTION(MONITOR) CRITERIA(00S) ENDED
RC=00
GEORPT * 16:35:25  STEP(013) MOP.KG.A6QS1 FUNCTION(GMSTOP) SITE(SITE1) STARTED
GEORPT * 16:35:2 GEORPT * 16:35:50  STEP(027) MOP.KP.A6PS1 FUNCTION(GDPS) COMMAND(STOPMIRROR)
ENDED RC=00
GEORPT * 16:35:50  STEP(028) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) STARTED
GEORPT * 16:35:50  STEP(028) MOP.KP.A6PS1 FUNCTION(PESTPATH) CGROUP(NO) ENDED
RC=00 GEORPT * 16:36:01  STEP(036) MOP.KG.A6QS1 FUNCTION(PESTPAIR) ENDED RC=00
GEORPT * 16:36:01  STEP(037) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(FIRSTPASS)
STARTED
GEORPT * 16:36:02      (037) ALL MONITORED VOLUMES HAVE NOW COMPLETED FIRSTPASS
GEORPT * 16:36:02  STEP(037) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(FIRSTPASS)
ENDED RC=00
GEORPT * 16:36:02  STEP(038) MOP.KG.A6QS1 FUNCTION(PSUSPEND) BYPASSED - NO
COPYONCE VOLUMES
GEORPT * 16:36:02  STEP(039) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANGE)
SITE(SITE1) STARTED
GEORPT * 16:36:03  STEP(039) MOP.KG.A6QS1 FUNCTION(SITE) ACTION(CHANGE)
PRINT LOG/TRACE UTILITY                                09/12/23                    58
SITE(SITE1) ENDED RC=00
GEORPT * 16:36:03  STEP(040) MOP.KG.A6QS1 FUNCTION(GMSTART) SITE(SITE1) STARTED
GEORPT * 16:36:06  STEP(040) MOP.KG.A6QS1 FUNCTION(GMSTART) SITE(SITE1) ENDED
RC=00
GEORPT * 16:36:06  STEP(041) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:36:06      REGIONB DR COPY RESTARTED
GEORPT * 16:36:06  STEP(041) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:36:06  STEP(042) MOP.KG.A6QS1 FUNCTION(PDELPAIR) STARTED
GEORPT * 16:36:09  STEP(042) MOP.KG.A6QS1 FUNCTION(PDELPAIR) ENDED RC=00
GEORPT * 16:36:09  STEP(043) USA.KR.A6RS4 FUNCTION(PDELPAIR) STARTED

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PROCEDURE execution Summary Report



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GEORPT * 16:35:23 STEP(011) MOP.KG.A6QS1 FUNCTION(MONITOR) CRITERIA(00S)
STARTED
GEORPT * 16:35:24      (011) ALL MONITORED VOLUMES HAVE EXCEEDED COPY TARGET
99.500 GEORPT * 16:36:11 STEP(043) USA.KR.A6RS4 FUNCTION(PDELPAIR) ENDED RC=00
GEORPT * 16:36:11 STEP(044) USA.KG.A6QS4 FUNCTION(PDELPAIR) STARTED
GEORPT * 16:36:13 STEP(044) USA.KG.A6QS4 FUNCTION(PDELPAIR) ENDED RC=00
GEORPT * 16:36:13 STEP(045) USA.KG.A6QS4 FUNCTION(HMTCTRL) STARTED
GEORPT * 16:36:14 STEP(045) USA.KG.A6QS4 FUNCTION(HMTCTRL) ENDED RC=00
GEORPT * 16:36:14 STEP(046) MOP.KP.A6PS1 FUNCTION(HMTCTRL) STARTED
GEORPT * 16:36:15 STEP(046) MOP.KP.A6PS1 FUNCTION(HMTCTRL) ENDED RC=00
GEORPT * 16:36:15 STEP(047) MOP.KP.A6PS1 FUNCTION(HMTCTRL) STARTED
GEORPT * 16:36:16 STEP(047) MOP.KP.A6PS1 FUNCTION(HMTCTRL) ENDED RC=00
GEORPT * 16:36:16 STEP(048) MOP.KG.A6QS1 FUNCTION(HMTCTRL) STARTED
GEORPT * 16:36:17 STEP(048) MOP.KG.A6QS1 FUNCTION(HMTCTRL) ENDED RC=00
GEORPT * 16:36:17 STEP(049) MOP.KG.A6QS1 FUNCTION(HMTCTRL) STARTED
GEORPT * 16:36:18 STEP(049) MOP.KG.A6QS1 FUNCTION(HMTCTRL) ENDED RC=00
GEORPT * 16:36:18 STEP(050) MOP.KG.A6QS1 FUNCTION(MESSAGE) STARTED
GEORPT * 16:36:18      MOP.ANTIGONE VOLUMES RESTORED TO AN MGM4SITE CASCADED
CONFIGURATION
GEORPT * 16:36:18 STEP(050) MOP.KG.A6QS1 FUNCTION(MESSAGE) ENDED RC=00
GEORPT * 16:36:18 PROCEDURE EXECUTION SUCCESSFUL RC=00
GEORPT
*****
GEORPT * - END OF PROCEDURE EXECUTION LOG
GEORPT
*****
```


Examples of the MGM 4-site Procedures available



Name	Purpose
MGM4SITE_CHECK_TOPOLOGY	Check the disk topology for the MGM 4-site configuration. It checks for the PPRC status of the primary disk in the application region and then establishes the PPRC paths required in the recovery region.
MGM4SITE_MOVEGM	This procedure will determine the current mirroring configuration (primary disk in RS1 or RS2) and then internally will schedule the appropriate procedure to move the GM session from RS1 to RS2 or vice versa and result in two copies being active in the solution.
MGM4SITE_PREPARE_MIRROR	This procedure is intended to be used after recovering the disks and the systems in the recovery region to re-establish replication back to the former application region, preparing to return home. This procedure will schedule the specific procedure based on the mirroring topology when the GM was recovered, either cascaded or multi-target with primary disk in RS1 or RS2.
MGM4SITE_RESTORE	This procedure is used to re-introduce the intermediate disk subsystem after using MOVEGM to move the GM session. This procedure will determine the current mirroring configuration and then will schedule the appropriate actions to restore the intermediate disk removed as part of MGM4SITE_MOVEGM. This procedure can also be used to switch from an MGM 4-site multi-target to a cascaded configuration.
MGM4SITE_SWAP	This procedure is intended to permanently swap the primary replication site in the application region, that is HyperSwap the PPRC configuration and move the GM session to the new secondary disks.
MGM4SITE_SWITCH_CAS_RS1	Perform a planned region switch to move the workload to the recovery region. This procedure is used when the primary disk before the switch is in RS1 and the MGM 4-site configuration is cascaded. It will result in a Global Copy relationship to be established between RS1 in the recovery region and RS1 in the application region and between RS1 and RS2 in the application region.
Note: Similar procedures exist to be used when running cascaded from RS2, or multi-target from RS1 or RS2.	